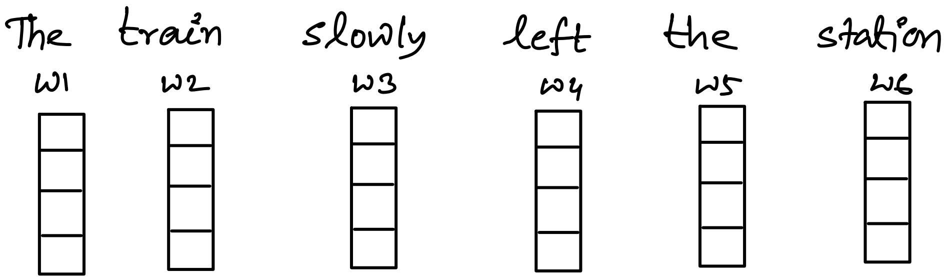
Transformer

- How to take the surrounding **context** of each word into account?
- How to take the order of the words into account?
- How to generate an output that has the same length as the input

① How to take the surrounding **context** of each word into account?

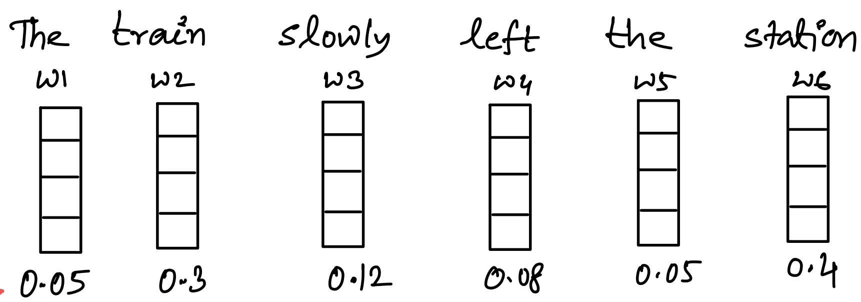


- How can we modify station's embedding so that it incorporates the other words?

→ Imagine that we somehow know how much attention to give the other words i.e., how much weights to give the other words

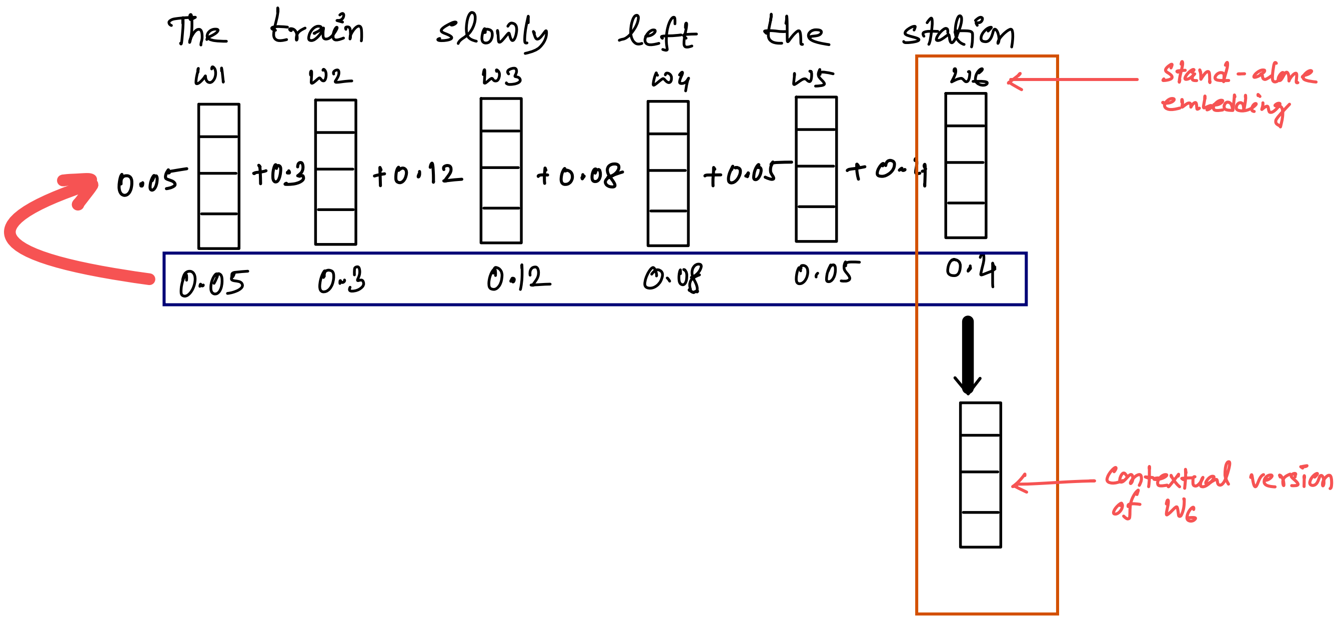
Intuitively:
Which word(s) should get the most weight, which word(s) the least?

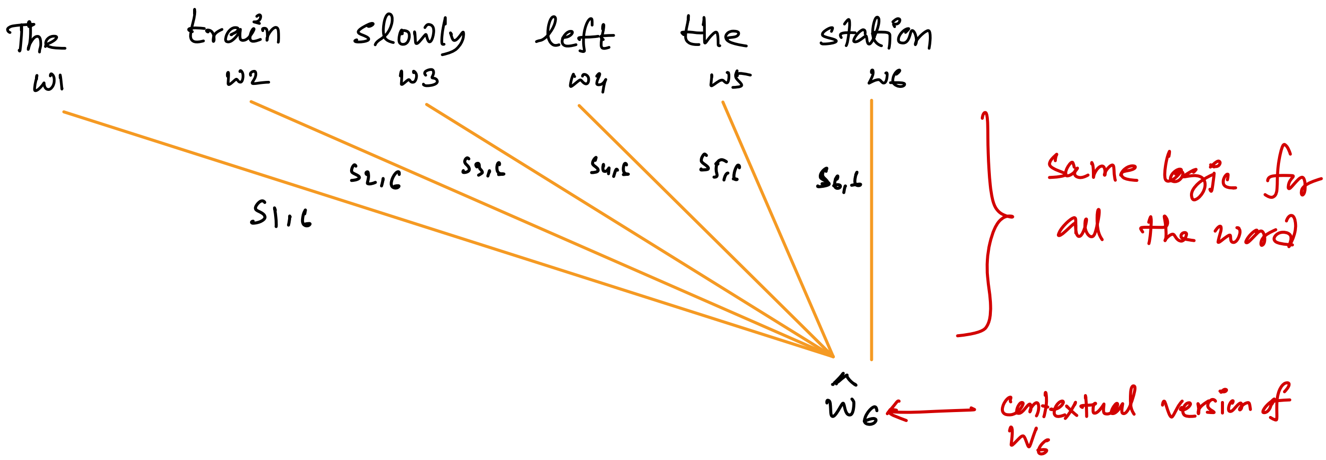
train ← lot of weight
slowly, left ← less weight



may be something like this?

- How can we use these weights to "Contextualize" the standalone embedding W_6 for "Station"

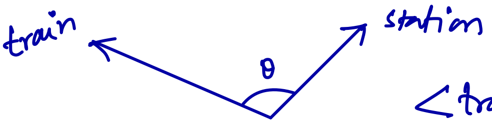




$$\hat{W}_6 = S_{1,6}W_1 + S_{2,6}W_2 + S_{3,6}W_3 + S_{4,6}W_4 + S_{5,6}W_5 + S_{6,6}W_6$$

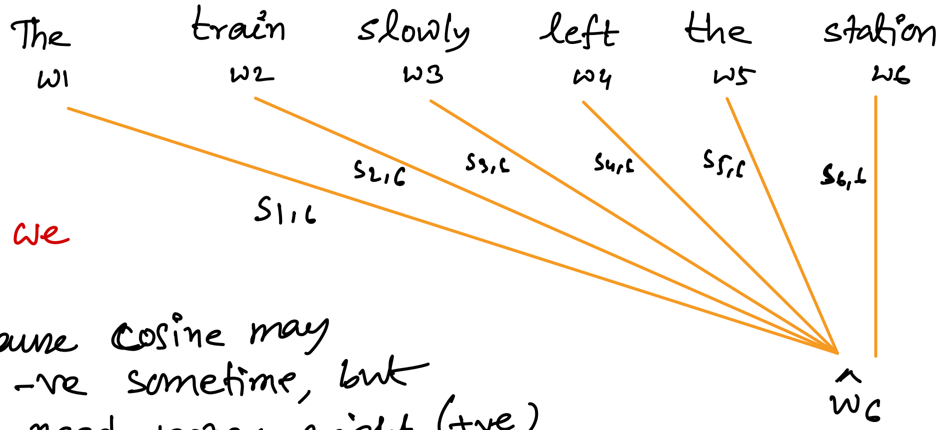
Intution:

- The weight of a word should be proportional to how related it is to the word "Station"
- One way to quantify how "related" two words are: the dot-product of their stand-alone embeddings



$\langle$train, station$\rangle$

= (length_train)$\cdot$(length_station) cos (angle)



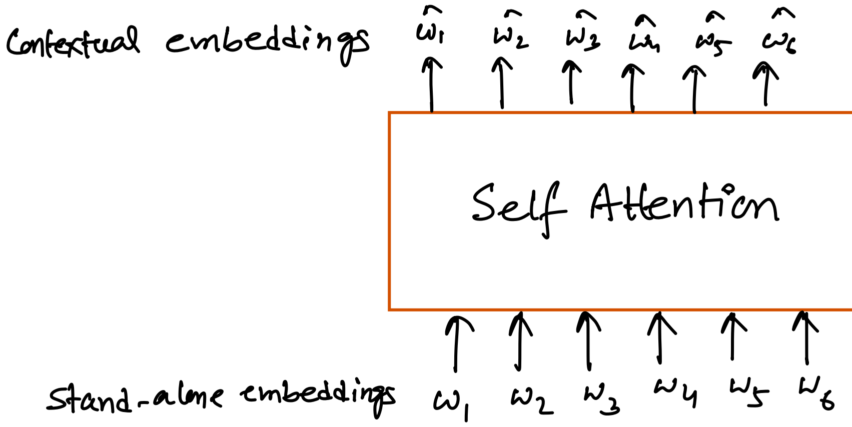
how should we do this?

→ because cosine may be -ve sometime, but we need proper weight (+ve)

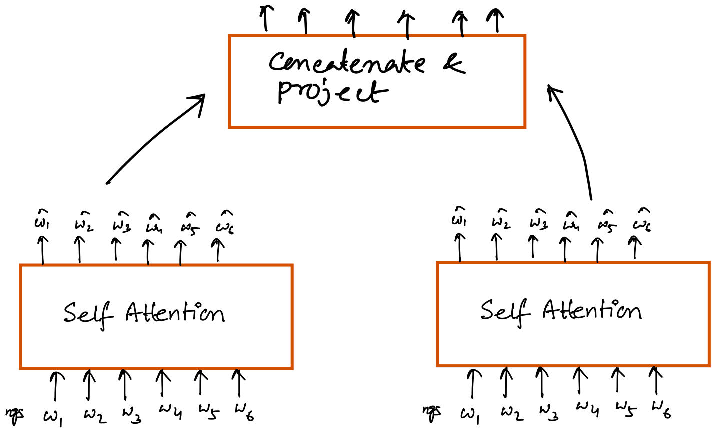
→ but sum of all should be 1

⟹ Softmax

$$S_{1,6} = \frac{\exp(w_1, w_6)}{\sum_{i=1}^{6} \exp(w_1, w_6)}$$

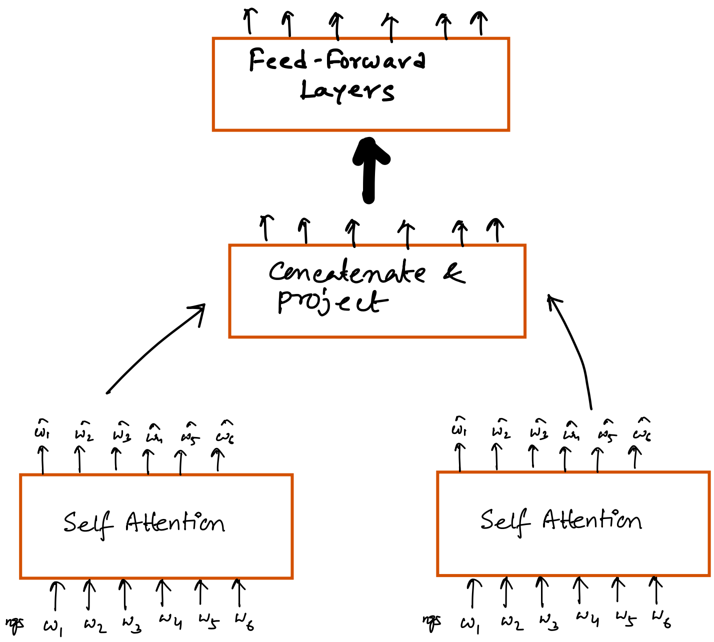


Multi-Head Attention:



- patterns related to "tense"
- patterns related to "tone"

....



Positional Encoding

- Stand-alone Embedding
  +
  positional Embedding

$\boxed{\text{GPT 3 has 96 Transformers}}$